

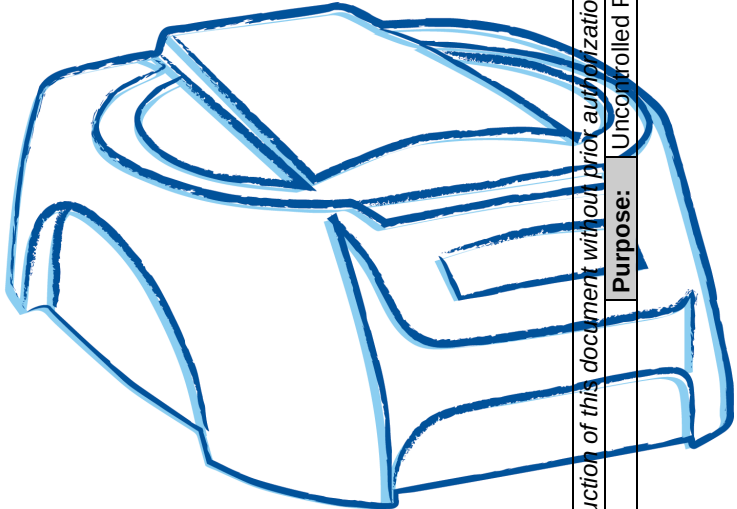
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DGspin

3044084
03/2016

DGspin
Instructions for Use



GRIFOLS

Reference:	REGD-0019335	Effective	Title:	3044084 Instrucciones de utilización DG SPIN (EN)	Version:	1.0; CURRENT; Most-Recent; Effective	Page	3 of 62
Status:	Effective						Effective Date:	07-Apr-2016

DG SPIN

Instructions for Use



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PLEASE READ THESE INSTRUCTIONS FOR USE CAREFULLY
BEFORE WORKING WITH THE DG SPIN CENTRIFUGE

Information about centrifuge safety can be found in Section 2.

The symbols used in these Instructions for Use are as follows:



WARNING: Indicates the possibility of personal injury, loss of life, or harm to the environment if the instructions are not followed.



CAUTION: Indicates the possibility of: 1) instrument damage if instructions are not followed; or 2) compromising of samples or sample integrity due to cross-contamination.



NOTE: Used for clarifications or additional or emphasized information.

The defined terms used in these Instructions for Use are as follows:

Operator:	The person operating the equipment for its intended use.
Supervisor (or Responsible Body):	The individual or group responsible for the use and maintenance of the equipment and for ensuring that operators are adequately trained.
Qualified Technician:	Person responsible for the installation, reparation and special maintenance of the equipment who has received specific training for this purpose.

All restrictions referred to Supervisor are valid for Operator.

Equipment design and specifications are subjected to change without previous notification.

This document is available in various languages. Translations have been carried out from the master document in English. In case of doubt or discrepancies, what is expressed in the master document in English will be given priority.

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DG SPIN

1 Introduction

THIS CENTRIFUGE IS INTENDED FOR
IN VITRO DIAGNOSTIC USE

These Instructions for Use are intended for the users of the DG SPIN Centrifuge and contain all the information necessary to safely and adequately work with the centrifuge.

Carefully read these Instructions for Use before you work with the centrifuge.

If you have any questions, please contact Technical Service before starting any operation.

These Instructions for Use should be accessible to all personnel who work with the centrifuge.

1.1 Intended Use

The DG SPIN Centrifuge is a device specifically designed to centrifuge DG Gel cards.

1.2 Principle of Operation

The DG SPIN Centrifuge is designed to centrifuge a maximum of 24 DG Gel cards in one run at 990 rpm for 9 minutes as indicated by the DG Gel test instructions. The centrifugation speed and time parameters have been set at the factory and are the parameters required to correctly centrifuge DG Gel cards.

1.3 Product Limitations

The DG SPIN Centrifuge is designed and manufactured only for the centrifugation of DG Gel cards and should not be used for any other type of test.

1.4 Operator Training

The use of the DG SPIN Centrifuge does not require specific training beyond these Instructions for Use and the training established by the laboratory.

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2 Safety Information

Grifols takes no responsibility for unauthorized modifications of the equipment or for failure of operating personnel to heed the warnings and cautions outlined in these Instructions for Use.

2.1 Warnings and Cautions

Before operating the equipment, review and become familiar with these warnings and cautions. Adherence to the warnings and cautions will help protect you, your colleagues, the equipment, the test results, and ultimately your patients.

2.1.1 Health and Safety Notices



WARNING: Use the centrifuge only for the operations described in these Instructions for Use.



WARNING: Unless specifically instructed to do so, do not drop or put anything into any opening in the centrifuge.



WARNING: Do not use the centrifuge if it is not working properly or if it is damaged. Examples of damage include:

- Damage caused by dropping the centrifuge.
- Damage caused by spilling liquid on or into the centrifuge.
- Damage caused by transport or storage under conditions outside of the transport or storage conditions listed in Section 3.1 or severe stresses during transport.
- Damage to the power cord or its plug.



WARNING: Do not use the centrifuge in the presence of flammable gases or volatile products. Do not put flammable, explosive, or chemical materials into the centrifuge.



WARNING: Do not lean on the centrifuge.



WARNING: Inappropriate use of this centrifuge can damage the safety features of this centrifuge.


WARNING: Chemical and Biological Hazards: Appropriate safety precautions, including Universal Precautions, must be strictly followed, as well as local, state, and national regulations. Appropriate personal protective equipment must be worn at all times when working with liquids in and around the equipment. Dispose of all chemical and biological waste, including empty sample and reagent containers or tubing, in accordance with local, state, and national regulations and practices. In the event of a spill of sample or reagent material, follow all site protocols for spill response, control, and disposal.


WARNING: Follow Universal Precautions when working with potentially hazardous materials, including the following circumstances:

- When handling samples. Treat these as potentially infectious material.
- When working with liquids in and around the equipment.
- When handling waste and waste containers. They may contain biohazardous waste.
- When performing maintenance activities.

 Always wear appropriate protective equipment when working with hazardous materials.


WARNING: Any part of the centrifuge that comes into contact with blood, serum samples, or any other biological liquid should be treated as potentially infectious.


CAUTION: Do not use accessories which are not supplied or recommended by the manufacturer.


CAUTION: The centrifuge may only be disassembled by qualified personnel.

2.1.2 Electrical Safety Notices


WARNING: To avoid risk of electric shock, this centrifuge should be connected to a grounded power outlet.


WARNING: To disconnect the centrifuge from the power supply, unplug the centrifuge (the power switch is not sufficient).


WARNING: Electromagnetic compatibility and electrical safety tests have been performed using the power cord supplied by the manufacturer. Do not use a power cord other than the power cord provided with the centrifuge as this may result in increased electromagnetic emissions and/or compromise the centrifuge.



WARNING: If you spill cleaning solution or other liquids into the centrifuge, the centrifuge must be unplugged, cleaned, and decontaminated. The centrifuge must only be disassembled by qualified personnel and it must not be plugged in again until it has been fully inspected by qualified personnel.

2.1.3 Installation Notices



WARNING: Do not let the centrifuge or its power cord come into contact with hot surfaces.



WARNING: Leave a space of at least 30 cm around the centrifuge. No person or any dangerous material should be within this area while the centrifuge is in operation.

2.1.4 Operational Notices



WARNING: During normal use of the DG SPIN Centrifuge, make sure that the security ring is protected by the security ring cover.



WARNING: Before opening the security ring cover, check that the centrifuge is unplugged and that the head is not moving.



WARNING: Opening the security ring cover immediately disconnects the motor. However, the head can still move. Before reaching into the centrifuge, make sure that the head has stopped moving.



WARNING: Always load an even number of DG Gel cards in the head and distribute them symmetrically so that the motor load is balanced. For example, load half of the cards starting from position 1 and the other half starting from position 13.



WARNING: Only qualified personnel should remove the security ring cover.



CAUTION: If centrifugation stops or is interrupted for any reason before the centrifugation process is complete, discard the cards and invalidate the results. Reliable results can only be obtained if the cards are subject to one complete and continuous centrifugation.



CAUTION: While the cards are being processed, do not dispense samples and/or reagents into the card holders.



CAUTION: Do not use a head other than those specified. Using an unauthorized head may cause deterioration and/or malfunction of the centrifuge.

- 
CAUTION: If the display shows an error message, restart the centrifuge. If the error persists, contact Technical Service and tell them the error code the centrifuge displayed.
- 
CAUTION: After a centrifugation batch has finished, another batch cannot be started until the cover has been opened. Open the cover and immediately remove the centrifuged cards to avoid centrifuging them twice.
- 
CAUTION: To handle DG Gel cards, follow the DG Gel Instructions for Use.
- 
CAUTION: Repairs should only be carried out by qualified personnel.
- 
CAUTION: To stop the alarm, press the **Open Cover** button , the **Start Centrifugation** button , or the **Stop Centrifugation** button . If the alarm continues, contact Technical Service.

2.1.5 Maintenance Notices

- 
WARNING: Before disassembling, cleaning, and/or decontaminating the centrifuge, unplug the centrifuge from the power supply.
- 
WARNING: During cleaning and/or decontamination, wear protective gloves.
- 
WARNING: Some cleaning or decontamination products, such as sodium hypochlorite solution, may be corrosive, irritating for the skin and eyes, or toxic if inhaled, absorbed, or ingested.
- 
CAUTION: Before disassembling, cleaning, and/or decontaminating the centrifuge, remove all cards and/or samples from inside the centrifuge.

2.1.6 Transport and Storage Notices

- 
WARNING: Decontaminate the centrifuge before transport and/or storage.
- 
CAUTION: Use only the original packaging to store or transport the centrifuge.

2.1.7 Disposal Notices



WARNING: Only qualified personnel should disassemble this centrifuge.



WARNING: The centrifuge contains electronics. Electronics can contain hazardous materials. Dispose of all electronics in accordance with local, state, and national regulations.

2.2 Centrifuge Labeling

The DG SPIN Centrifuge comes with the following labels:

- Packaging Label.
- Product Label.
- Biological Risk Level Label.

The following symbols are used on the labels:



Serial number



Catalog number



Temperature range permitted during storage and transport



Permitted relative humidity range during storage and transport



In vitro diagnostic medical device



Manufacturer



Consult Instructions for Use



Biological risk, infectious substance



Date of manufacture



DataMatrix including the following information:

- (01) GTIN number
- (11) Manufacturing year
- (21) Serial number of the instrument



Safety mark that certifies that the centrifuge meets the applicable USA and Canadian safety standards.

Packaging Label

The Packaging Label (Figure 1) is located on the right upper part of the front of the box for the DG SPIN Centrifuge.

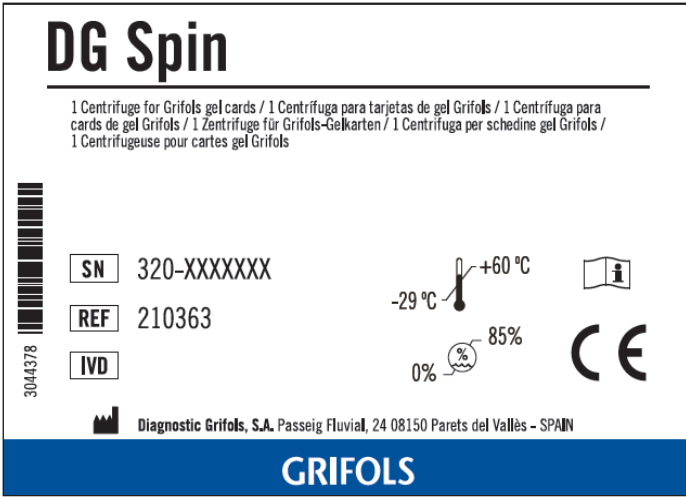


Figure 1. Packaging Label

Product Label

The Product Label (Figure 2) is located on the rear of the centrifuge and contains technical information (voltage, frequency, input power, and fuse specifications), as well as the model name, catalogue number, serial number, manufacturing date, name and address of the manufacturer, and regulatory information.

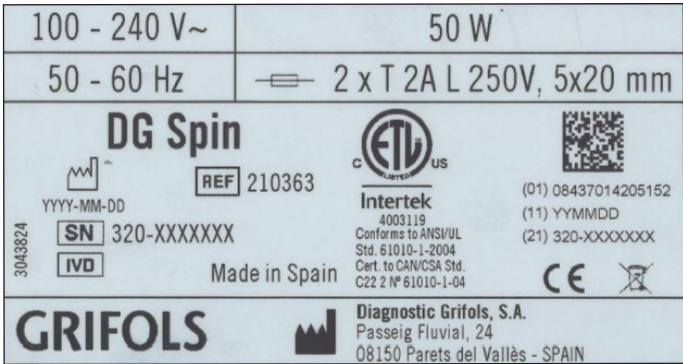


Figure 2. Product Label

Biological Risk Level Label

The Biological Risk Level Label (Figure 3) is located on the side of the centrifugation chamber and warns the operator about potential biological risks caused by residual substances deposited on the surface.



Figure 3. Biological Risk Level Label

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DG SPIN

3 Technical Specifications and Accessories

The following sections contain information on the technical specifications and accessories available for the DG SPIN Centrifuge.

3.1 Technical Specifications

Table 1 lists the technical specifications for the centrifuge.

Table 1. Technical Specifications

Feature	Technical Specifications	
MODEL	DG SPIN	
LOAD CAPACITY	24 DG Gel cards	
DG SPIN HEAD	Removable and interchangeable	
CENTRIFUGATION CONDITIONS	Pre-programmed for DG Gel cards only	
CENTRIFUGATION SPEED	990 rpm $\pm$ 10 rpm for DG Gel cards only	
CENTRIFUGATION SPEED UNITS DISPLAYED	Programmable, rpm or g	
CENTRIFUGATION TIME	9 minutes	
ALARM	Yes, to indicate end of centrifugation and error situations	
VOLUME OF ALARM	Programmable, 3 levels	
SAFETY SYSTEMS	- Out-of-balance control. - Detection of door opening. - Safety closure.	
INTERFACE	LCD screen and buttons	
POWER REQUIREMENTS	Voltage:	100-240 V ~
	Frequency:	50-60 Hz
	Input power:	50 W
	Fuses:	2 x T 2A L 250 V, 5 x 20 mm

Feature	Technical Specifications	
PROTECTION AGAINST ELECTRIC SHOCK	Class:	I
INSTALLATION CATEGORY	Overvoltage category II (local level, appliances, portable equipment, etc.)	
DIMENSIONS	525 mm (depth) x 410 mm (width) x 180 mm (height)	
WEIGHT	Approximately 9 kg	
OPERATING CONDITIONS	Indoor use	
	Temperature:	15 °C to 40 °C
	Maximum relative humidity non-condensing:	80%
	Maximum altitude:	9843 ft (3000 m)
	Power supply maximum voltage fluctuations:	±10% of nominal voltage
TRANSPORT AND STORAGE CONDITIONS	Temperature:	-29 °C to 60 °C
	Maximum relative humidity non-condensing:	85%
USEFUL LIFE	5 years (based on an average use of 8 hours/day and 200 days/year)	

3.2 Accessories

Table 2 lists the accessories available for the centrifuge.

Table 2. Accessories

PART #	DESCRIPTION	QUANTITY
210364	DG SPIN Head	1

4 Description

Figures 4 and 5 respectively show a front and rear view of the DG SPIN Centrifuge.

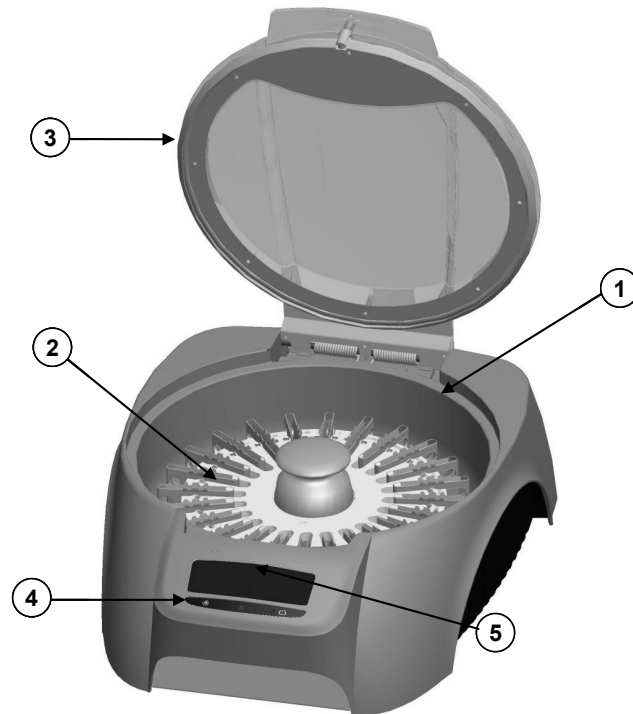


Figure 4. Front View of the DG SPIN Centrifuge

- (1) Centrifugation chamber
- (2) DG SPIN Head
- (3) Centrifuge cover
- (4) Control buttons
- (5) Display

Description

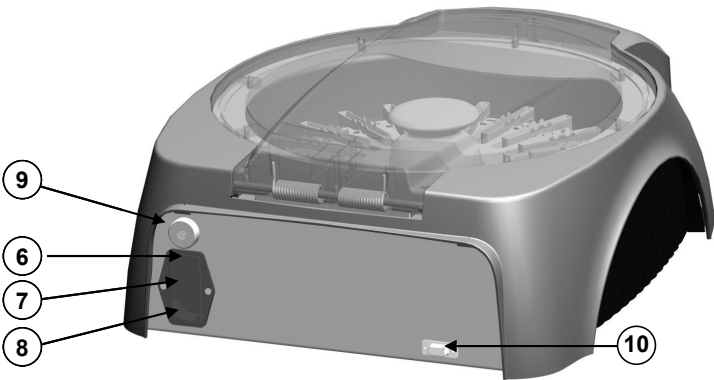


Figure 5. Rear View of the DG SPIN Centrifuge

- (6) Power switch
- (7) Fuse housing
- (8) Power cord connection
- (9) Security ring cover
- (10) Computer connection (for the exclusive use of Technical Service)

4.1 Centrifugation Chamber

The centrifugation chamber (Figure 6, no. 1) houses the removable head. The chamber helps to contain any spills or leaks to make cleaning easier. The centrifugation chamber also has a cover (Figure 6, no. 3) that protects the motor.

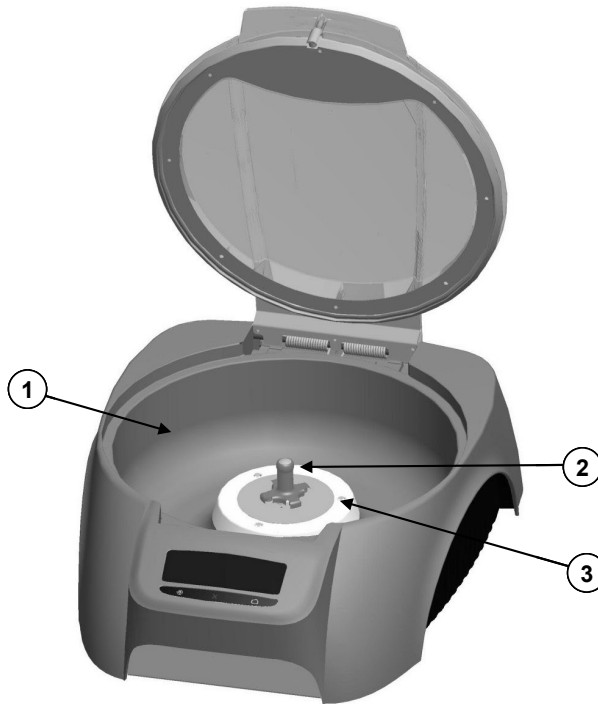


Figure 6. Centrifugation Chamber

- (1) Centrifugation chamber
- (2) Lean axle and CrossHead of head
- (3) Motor cover

4.2 DG SPIN Head

The head (Figure 7) is a circular support with 24 numbered positions for card holders (Figure 7, no. 2) to hold the DG Gel cards for centrifugation.

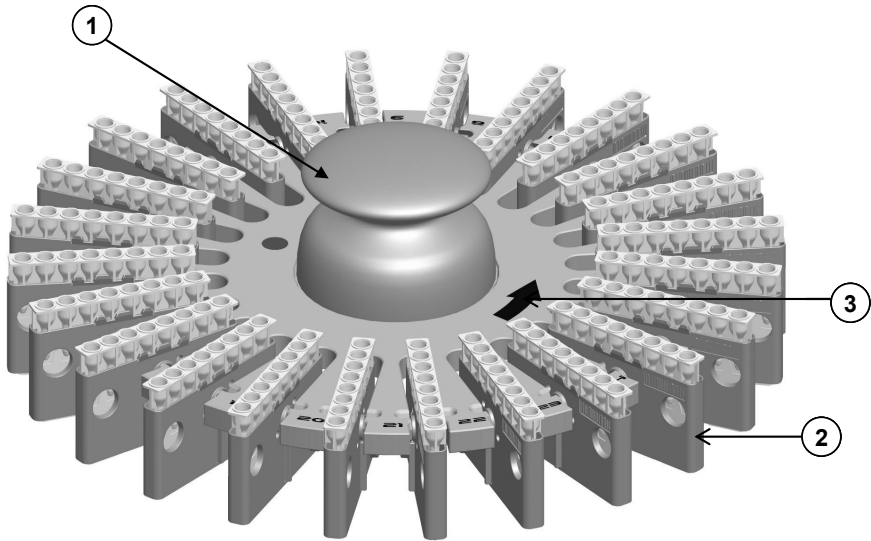


Figure 7. DG SPIN Head

- (1) Knob
- (2) DG Gel card holders
- (3) Direction of centrifugation

The head is removable to make it easier for the operator to load and unload the cards. To remove the head, lift it vertically with the knob (Figure 7, no. 1) until it is released from the lean axle (Figure 6, no. 2).

To load the DG SPIN Head onto the centrifuge, the crosshead of the lean axle is magnetized to help facilitate correct positioning of the head.

4.3 Centrifuge Cover

The DG SPIN Centrifuge has a transparent cover (Figure 4, no. 3) with a hinge and an electromagnetic safety closure.

4.4 Control Buttons

The DG SPIN Centrifuge has three buttons in the area below the display (Figure 8) to control centrifugation.

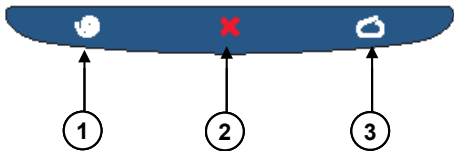


Figure 8. Centrifugation Control Buttons

- (1) **Start Centrifugation** button 
- (2) **Stop Centrifugation** button 
- (3) **Open Cover** button 

4.4.1 Start Centrifugation Button

After making sure that the cover is closed, press the **Start Centrifugation** button  to start the centrifugation cycle.

4.4.2 Stop Centrifugation Button

Press the **Stop Centrifugation** button  to stop centrifugation before the centrifugation time ends (the alarm will sound). Press and hold the **Stop Centrifugation** button  for several seconds to stop the alarm.

4.4.3 Open Cover Button

Press the **Open Cover** button  to open the cover (Figure 4, no. 3). The cover can only be opened once the motor has stopped completely.

4.5 Display

The display (Figure 9) shows the time remaining to the end of centrifugation and the centrifugation speed. The display also shows possible error symbols.

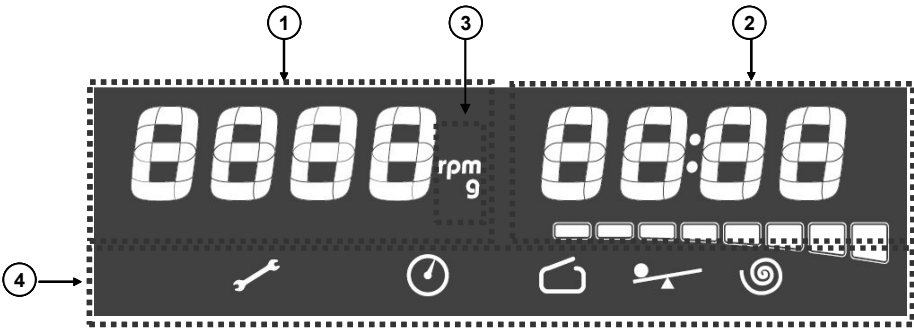


Figure 9. Display

- (1) Centrifugation speed information
- (2) Timer information
- (3) Centrifugation speed units
- (4) Errors area

4.5.1 Remaining Centrifugation Time Information

In normal mode, the display shows the remaining time for centrifugation with a digital counter (Figure 10, no. 1) and a progress bar (Figure 10, no. 2). The progress bar fills in as the pre-programmed time elapses.

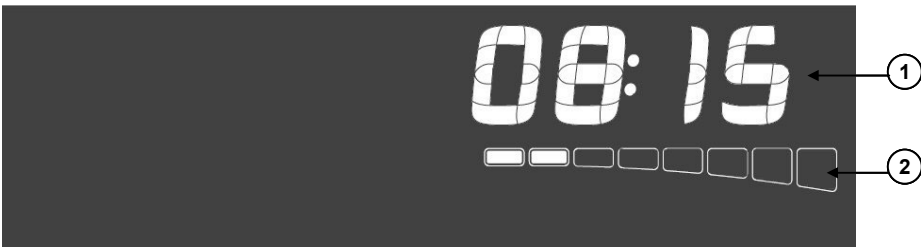


Figure 10. Display During Centrifugation

- (1) Remaining centrifugation time
- (2) Progress bar

4.5.2 Centrifugation Speed Units

The DG SPIN displays centrifugation speed in two formats: “rpm” or “g.” To change the centrifugation speed format, see Section 5.4.

4.5.3 Error Symbols Area

The error symbols area is located on the left lower part of the display (Figure 9, no. 4) and displays symbols for various error situations that occur during start-up or centrifugation. For more information, see Section 12.1.

The following symbols may be displayed in the error symbols area:



Repair symbol: Contact Technical Service



Erroneous Centrifugation Time symbol: Batch completed with erroneous centrifugation time



Cover Open symbol: The centrifuge cover is open



Out-of-Balance symbol: Out-of-balance during centrifugation



Speed Out-of-Range symbol: Centrifugation speed out-of-range

4.6 Security Ring Cover

Under normal conditions, the DG SPIN Centrifuge will open when the **Open Cover** button  is pressed. However, if you need to open the DG SPIN when the power goes out or the electromagnetic security closure malfunctions, the DG SPIN Centrifuge has a security ring cover in the back of the centrifuge (Figure 5, no. 9).

Description

- **WARNING:** Only qualified personnel should remove the security ring cover.
- **WARNING:** During normal use of the DG SPIN Centrifuge, make sure that the security ring is protected by the security ring cover.
- **WARNING:** Opening the security ring cover immediately disconnects the motor. However, the head can still move. Before reaching into the centrifuge, make sure that the centrifuge is unplugged and that the head has stopped moving.

To remove the security ring cover, unscrew the screws that secure the security ring cover and remove the cover.

5 Installation

The following sections contain instructions on installing the DG SPIN Centrifuge, as well as information about changing the settings.

5.1 Installation Requirements



WARNING: Leave a space of at least 12 in (30 cm) around the centrifuge. No person or any dangerous material should be within this area while the centrifuge is in operation.



WARNING: Do not let the centrifuge or its power cord come into contact with hot surfaces.



WARNING: To avoid risk of electric shock, this centrifuge should be connected to a grounded power outlet.



CAUTION: Do not place the centrifuge outdoors or where it could get wet.



NOTE: If the centrifuge has been stored in environmental conditions different from operating conditions, keep the centrifuge in operating conditions for at least one hour before plugging it in. For more information on operating conditions, see Section 3.1.

The installation requirements for the DG SPIN are as follows:

- A minimum space of 50 cm wide by 60 cm deep (includes space to connect the power cord) by 56 cm high.
- Do not position the centrifuge so that it is difficult to operate the power switch (Figure 5, no. 6).
- Do not place anything on top of the centrifuge.
- Connect the centrifuge to a power supply which meets the power supply requirements listed in Section 3.1 (including ground protection).
- Install the DG SPIN Centrifuge on a flat, horizontal, and stable surface.

5.2 Unpacking the Centrifuge

After you choose an appropriate location to install the centrifuge, unpack the centrifuge by following these instructions:

1. Open the upper part of the box.
2. Remove all packing protection and accessories.
3. Remove the protection bag and place the centrifuge in the chosen position.

4. Make sure that all elements specified in the attached *Packing List* are included.
5. Check that the centrifuge has not been damaged during storage or transport.

5.3 Installing the Centrifuge

To install the DG SPIN Centrifuge, follow these instructions:

1. Connect the power cord to the centrifuge (Figure 5, no. 8) and to a grounded outlet.
2. Turn on the power switch (Figure 5, no. 6).
3. Confirm that the display shows the pre-programmed centrifugation time of nine minutes.
4. Press the **Open Cover** button  and remove the internal protection that protects the head.
5. Remove the head from the centrifugation chamber and make sure that all the card holders (Figure 7, no. 2) are placed correctly and held by the corresponding pivot.
6. Close the cover by pressing the front of the cover.
7. Make sure that the cover is completely closed and press the **Start Centrifugation** button .
8. Make sure that the head starts to rotate.
9. After five minutes, press the **Start Centrifugation** button  again and make sure that the centrifugation speed displayed is within the speed range listed in Section 3.1.
10. Stop centrifugation by pressing and holding the **Stop Centrifugation** button  until the alarm emits several short beeps followed by a long beep.
11. Wait for 30 seconds and make sure that the head has stopped and that the display shows the pre-programmed centrifugation time of nine minutes.

The DG SPIN Centrifuge is supplied with a conformity certificate stating that the centrifugation speed (990 rpm) has been verified using instruments calibrated by accredited entities. The centrifugation speed does not need to be recalibrated during its useful life if it is used under normal working conditions.

5.4 Changing Settings

The settings for the DG SPIN Centrifuge have been preset to the following values:

- Alarm volume: High.
- Centrifugation speed units: rpm.
- Date and time: Greenwich Mean Time (GMT) time zone.



NOTE: Before changing settings, check with your supervisor.



NOTE: The Date and Time setting is used to maintain traceability in a “log file” with relevant information and possible incidences that have occurred during centrifugation. Only Technical Service personnel can access and use this file.

To change these settings, follow these instructions:

1. Turn the power off at the power switch.
2. Press and hold the **Start Centrifugation** button  and turn on the power at the power switch (Figure 5, no. 6).

The DG SPIN Centrifuge starts in **Configuration Mode** and the display shows the alarm volume menu. Three settings are available:
 - VOL 0: Deactivated.
 - VOL 1: Medium volume.
 - VOL 2: High volume.
3. Press the **Open Cover** button  to change the volume setting to the desired level.
4. Press the **Start Centrifugation** button  to move to the centrifugation speed units menu.
5. Press the **Open Cover** button  to change the centrifugation speed units setting. Two settings are available:
 - SEL rpm: Revolutions per minute (rpm).
 - SEL g's: Relative centrifugation force (g).
6. Press the **Start Centrifugation** button  to move to the date and time menu. The Display flashes the date (YYYY.MM.DD) and time (HH:MM).
7. Press the **Open Cover** button  to modify the value of each field and then press the **Start Centrifugation** button  to move to the next field.



NOTE: The date and time are preset to GMT.

8. Save the settings by pressing and holding the **Start Centrifugation** button  until the alarm emits a long beep.

The display shows the pre-programmed centrifugation time of nine minutes.

Installation



NOTE: If no button is pressed for 15 seconds at any point while settings are being changed, the centrifuge leaves **Configuration Mode** and the changes to the settings are not saved.



NOTE: To save changes to the settings, press and hold the **Start Centrifugation** button  until the alarm emits a long beep. Otherwise, the settings will not be saved.

6 Operation

The following sections contain information about the operation of the DG SPIN Centrifuge.



WARNING: Do not use the centrifuge if it is not working properly or if it is damaged. Examples of damage include:

- Damage caused by dropping the centrifuge.
- Damage caused by spilling liquid on or into the centrifuge.
- Damage caused by transport or storage under conditions outside of the transport or storage conditions listed in Section 3.1 or severe stresses during transport.
- Damage to the power cord or its plug.

6.1 Processing a Centrifugation Batch

To correctly use the DG SPIN Centrifuge, follow these instructions:

1. Turn the power on with the power switch (Figure 5, no. 6) and make sure that the display shows the pre-programmed centrifugation time of nine minutes.
2. Open the cover by pressing the **Open Cover** button  and remove the head by pulling it up until it is released from the lean axle.
3. Place the head on a flat surface.
4. Load the DG Gel cards in the card holders of the head.



WARNING: Always load an even number of DG Gel cards in the head and distribute them symmetrically so that the motor load is balanced. For example, load half of the cards starting from position 1 and the other half starting from position 13.



CAUTION: Do not use a head other than those specified. Using an unauthorized head may cause deterioration and/or malfunction of the centrifuge.



NOTE: If you have a second DG SPIN head (available as an accessory under Part # 210364), you can load one head with cards while the other head is being centrifuged.

To use an additional head, no special installation or configuration is necessary.

5. After the cards have been loaded in the head, place the head inside the centrifuge.
6. Close the cover by pressing the middle front part of the cover.
7. Press the **Start Centrifugation** button  to start the centrifugation batch. The timer is activated.

- 
CAUTION: While the cards are being processed, do not dispense samples and/or reagents into the card holders.
- 
CAUTION: The direction of centrifugation is important to standardize the work and make centrifugation conditions of the card uniform.

Consequently, place the DG Gel cards in such a way that the barcode coincides with the barcode symbolized in the card holder as shown by Figure 11.



Figure 11. Cards Orientation in the Card Holders

- After centrifugation has finished, the head stops automatically, the timer flashes, and the alarm beeps.
- Press the **Open Cover** button  to open the cover.
- Immediately remove the head.

- 
CAUTION: After a centrifugation batch has finished, another batch cannot be started until the cover has been opened. Open the cover and immediately remove the centrifuged cards to avoid centrifuging them twice.

To carry out a new centrifugation batch, load the head with new cards and close the cover. The timer will automatically show the pre-programmed centrifugation time of nine minutes.

6.2 Displaying the Centrifugation Speed

To momentarily view the centrifugation speed (or the force of the centrifugation, if the centrifugation speed unit setting is set to "g") (as shown in Figure 12), press any button during the centrifugation batch.

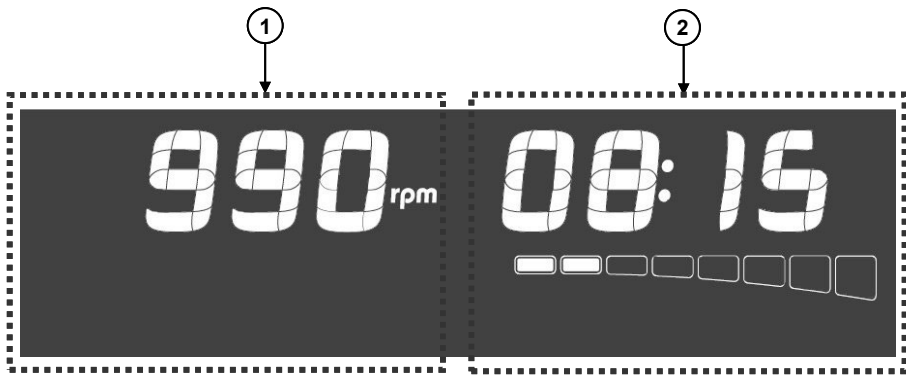


Figure 12. Centrifugation Speed and Timer Information on the Display

- (1) Centrifugation speed or force
- (2) Timer information

6.3 Stopping Centrifugation

To stop centrifugation before the centrifugation time is finished:

- Press and hold the **Stop Centrifugation** button  for a few seconds until the alarm emits several short beeps followed by a long beep.

The Display alternates between the time remaining when centrifugation stopped and the

Erroneous Centrifugation Time symbol .

- Press any button to silence the alarm.
- To carry out a new centrifugation batch, open the cover by pressing the **Open Cover** button .
- Load the head with a new set of cards.
- Close the cover.

The **Erroneous Centrifugation Time** symbol  disappears and the timer shows the pre-programmed centrifugation time of nine minutes.



CAUTION: If centrifugation stops or is interrupted for any reason before the centrifugation process is complete, discard the cards and invalidate the results. Reliable results can only be obtained if the cards are subject to one complete and continuous centrifugation.

7 Maintenance

During the useful life under average conditions (as listed in Section 3.1), maintenance of the centrifuge is limited to regular cleaning and decontamination. After the useful life period, the centrifuge can still be used but, in addition to regular cleaning and decontamination, regular (at least yearly) and functional inspections are recommended to confirm that the centrifugation speed is within the required range, that the timer is accurate, and that the out-of-balance control system is operating correctly. Perform these inspections by following the procedures listed in Section 8.

7.1 Maintenance Plan

The maintenance plan is a set of operations carried out at regular intervals to preserve the function of the DG SPIN Centrifuge. As a general rule, and as long as there is no irregularity in the operation of the centrifuge, follow the maintenance plan listed in Table 3 to ensure the proper functioning of the centrifuge.

Table 3. Maintenance Schedule

FREQUENCY	ACTIVITY	CARRIED OUT BY
Whenever samples or other substances are spilled in the centrifugation chamber or on the head	Clean the chamber and the head as described in Section 7.2.	OPERATOR
Monthly	Decontaminate the centrifuge as described in Section 7.3.	OPERATOR
Monthly	Confirm the integrity of the head and card holders as described in Section 7.4.	OPERATOR

Decontamination operations carried out by an operator or any other maintenance or repairs carried out by qualified personnel should be noted in Appendix A (included at the end of these Instructions for Use).

7.2 Cleaning the Centrifuge



WARNING: During cleaning and/or decontamination, wear protective gloves.

- 
WARNING: Before disassembling, cleaning, and/or decontaminating the centrifuge, unplug the centrifuge from the power supply.
- 
CAUTION: Before disassembling, cleaning, and/or decontaminating the centrifuge, remove all cards and/or samples from inside the centrifuge.
- 
CAUTION: If any liquid is spilled into the centrifuge, the centrifuge must be unplugged, cleaned, and decontaminated. The centrifuge must only be disassembled by qualified personnel and it must not be plugged in again until it has been fully inspected by qualified personnel.

Biological samples, saline solutions, acid, or alkaline solvents can cause damage to the exterior surface of the centrifuge and should be removed immediately. Periodically wipe the exterior surfaces with a cloth dampened with a mild detergent.

7.3 Decontaminating the Centrifuge

- 
WARNING: Any part of the centrifuge that comes into contact with blood, serum samples, or any other biological liquid should be treated as potentially infectious.
- 
WARNING: During cleaning and/or decontamination, wear protective gloves.
- 
WARNING: Before disassembling, cleaning, and/or decontaminating the centrifuge, unplug the centrifuge from the power supply.
- 
CAUTION: Before disassembling, cleaning, and/or decontaminating the centrifuge, remove all cards and/or samples from inside the centrifuge.

To prevent risk of infection, decontaminate the centrifuge monthly, as well as in the following situations:

- After spills or leaks of potentially contaminating liquids.
- Before and after transport or storage.
- Before repairs are performed.
- Before centrifuge disposal.

To decontaminate the centrifuge, follow these instructions:

1. Open the cover by pressing the **Open Cover** button .
2. Remove the head.
3. Turn off the centrifuge and disconnect the power cord.
4. In case of spills, wipe up the liquid with a disposable absorbent material (for example, paper towels, gauze, or cloth).

- Wipe the entire centrifuge (chamber, cover, head, card holders, etc.) with disposable towels or cotton balls dipped in a 0.5% (v/v) aqueous solution of sodium hypochlorite until all the surfaces of the centrifuge are completely damp. Any other solution that is used must be validated by the individual site. The disposable towels or cotton balls should be dipped in the disinfectant solution until they are completely damp.
- Rinse the surface with a soft detergent and water to eliminate odors and any residual solution.
- Dry the surface.
- Dispose of contaminated materials used during decontamination in a container for biological waste.



NOTE: The procedures described above do not guarantee that the centrifuge is completely decontaminated, but these procedures reduce the risk of contamination to a minimum.

7.3.1 Handling Decontamination Solution



WARNING: Some cleaning or decontamination products, such as sodium hypochlorite solution, may be corrosive, irritating for the skin and eyes, or toxic if inhaled, absorbed, or ingested.

During handling of the decontamination solution, follow these recommendations:

- These Instructions for Use.
- The laboratory’s safety regulations.
- Any applicable local, state, or national regulations.

7.4 Confirming the Integrity of the DG SPIN Head and Card Holders

To confirm the integrity of the DG SPIN head and card holders, follow these instructions:

- Open the cover by pressing the **Open Cover** button .
- Remove the head.

Inspect the head and the card holders for damage (i.e., cracks, bumps, or other visible damage). Do not use the head or the card holders if either of them has suffered any damage. If you find any damage, use a new head or card holder.

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8 Confirming Specifications (Operation Qualification Procedure)

The centrifugation speed of the DG SPIN Centrifuge has been calibrated by the manufacturer and centrifugation timing is controlled by two non-adjustable independent timers. If no function alarm is activated (see Sections 12.1 and 12.2), it is warranted that the centrifuge will centrifuge correctly for its useful life (see Table 1 in Section 3.1) without any additional calibration.

However, laboratories may need to periodically confirm that the centrifuge meets its specifications due to a quality management system or because they wish to use the centrifuge beyond its recommended useful life. To carry out these procedures, follow the instructions in Sections 8.2, 8.3, and 8.4 to confirm the following parameters:

- Centrifugation speed.
- Centrifugation timer.
- Out-of-balance control system operation.

8.1 Required Materials

To confirm centrifugation speed, the timer, and the correct operation of the out-of-balance control system, the following materials are required:

- Calibrated optical tachometer.
- Calibrated chronometer (precision: ± 1 second).
- 24 DG Gel cards.

Calibrate the optical tachometer using instruments controlled with traceable materials at primary or secondary reference standards.



NOTE: Any tachometer that meets the following specifications can be used:

- Type: Optical (white light).
- Scale division (Resolution): ≤ 0.1 rpm.
- Minimum range: 5-9999 rpm.
- Maximum variation permitted: ± 3 rpm (at 990 rpm).
- Reading distance: ≥ 6 in (15 cm).

8.2 Confirming Centrifugation Speed

Confirm that the DG SPIN Centrifuge complies with the speed specifications described in Section 3.1 by following these instructions:

1. Turn on the centrifuge with the power switch (Figure 5, no. 6).

- Confirm that the display is switched on and that the timer indicates the pre-programmed centrifugation time of nine minutes.
- Open the cover and remove the head.
- Load 24 DG Gel cards into the card holders.
- Load the head onto the centrifuge and close the cover.
- Press the **Start Centrifugation** button  to start a centrifugation batch.
- Wait 20 seconds (to ensure stabilization of centrifugation speed), and then hold the optical tachometer vertically and point it through the centrifuge cover toward the reflective mark of the head (Figure 13, no. 1).

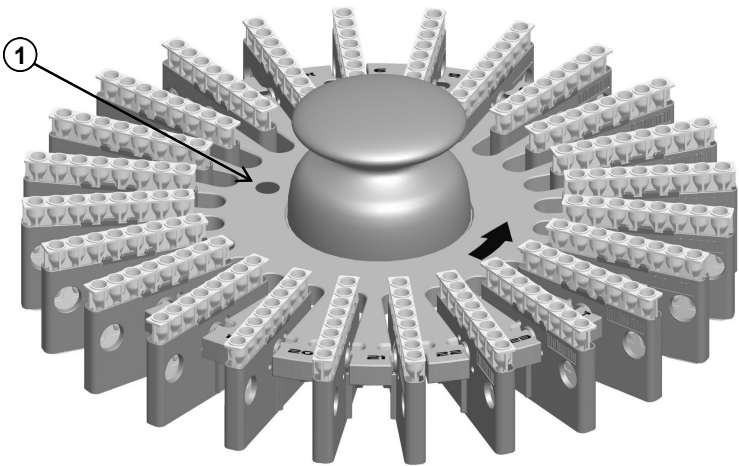


Figure 13. Reflective Mark for Measuring Centrifugation Speed

- Reflective mark
- Confirm that the speed shown on the tachometer is within the speed range listed in Section 3.1.
- Confirm that the speed shown by the display is also within the speed range listed in Section 3.1.



NOTE: Make sure that the centrifugation speed units display setting is set to “rpm.” To change the centrifugation speed units setting, see Section 5.4.



NOTE: The speed shown on the tachometer and the speed displayed by the centrifuge must be within 10 rpm of each other.

10. Stop centrifugation by pressing and holding the **Stop Centrifugation** button  until the alarm emits a long beep. Open the cover of the centrifuge, and remove the head and the cards.



NOTE: If the centrifugation speed measurement is not within the specifications listed in Section 3.1, repeat the test. If the results of the new test are not within the specifications, contact Technical Service.

8.3 Confirming the Timer

To confirm that the timer functions correctly:

1. Turn on the centrifuge with the power switch (Figure 5, no. 6).
2. Ensure that the timer shows the pre-programmed centrifugation time of nine minutes.
3. Press the **Start Centrifugation** button  to start a centrifugation batch and start the chronometer at the same time.
4. Stop the chronometer at the same time as the centrifugation timer finishes. Confirm that the difference in the time between the centrifuge timer and the chronometer is less than 10 seconds.



NOTE: If the timer is not within 10 seconds of the chronometer timing, repeat the test. If the results of the new test are not within the specifications, contact Technical Service.

8.4 Confirming Out-of-Balance Control System Operation

The out-of-balance control system stops centrifugation if an imbalance occurs in the head during centrifugation. This system also has an auto-verification system; if the out-of-balance control system fails, the centrifuge stops and the display shows the **Erroneous Centrifugation Time**

symbol , the **Out-of-Balance** symbol , and the **Repair** symbol .



NOTE: Perform the test of the Out-of-Balance Control System with the centrifuge on a flat, stable surface.

To confirm that this out-of-balance control system is working correctly, follow these instructions:

1. Open the cover and load two cards next to each other.
2. Close the cover.

3. Start a centrifugation batch by pressing the **Start Centrifugation** button .
4. Confirm that centrifugation stops and that the **Out-of-Balance** symbol  is displayed.



NOTE: If centrifugation does not stop and/or the **Out-of-Balance** symbol  is not displayed, contact Technical Service.

9 Transport and Storage



WARNING: Decontaminate the centrifuge before transport and/or storage.



CAUTION: Use only the original packaging to store or transport the centrifuge.

If you need to store the centrifuge for a long period of time, pack the centrifuge in the same manner as it was packed for transportation. Store the centrifuge in environmental conditions that meet the requirements listed in Section 3.1.

To store the centrifuge, you need a space with the following specifications:

- Minimum dimensions: 32 cm high by 64 cm deep by 53 cm wide (with packaging).
- Load-bearing capacity: 31 lbs (14 kg).

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10 Disposing of the Centrifuge

To dispose of the DG SPIN Centrifuge, remove all remains of samples and reagents, and then clean and decontaminate it. Upon completion, dispose of the centrifuge as electronic waste, according to local, state, and national regulations.

If you have any questions about the disposal of the centrifuge, contact Technical Service.



WARNING: The centrifuge contains electronics. Electronics can contain hazardous materials. Dispose of all electronics in accordance with local, state, and national regulations.

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11 Warranty

The warranty conditions are established in separate documents. For more information, contact Technical Service.

The warranty does not cover:

- Damage caused by using the centrifuge in environmental conditions outside of the conditions listed in Section 3.1.
- Damage caused by accident, negligence, or breach of these Instructions for Use.
- Damage caused for reasons beyond our control (atmospheric, geological phenomena, etc.).
- Damage caused by incorrect handling, including handling by unauthorized personnel.
- The plastic or rubber components, as well as the enamel or paint, that have deteriorated due to normal usage, except in the case of a manufacturer defect.
- Damage to the head, except in the case of a manufacturer defect.
- Damage that occurs during transport.

The replacement of equipment or components will be subject to the criteria of Diagnostic Grifols S.A.



CAUTION: Repairs should only be carried out by qualified personnel.

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12 Errors and Alarms

An operator can resolve most of the error messages and alarms that may occur during normal operation of the DG SPIN Centrifuge. The following sections describe possible error situations and the steps to resolve them, as well as the situations that might activate the alarm.

12.1 Clearing Errors

Table 4 lists the error symbols that the centrifuge displays in certain error situations, along with a description of the error and the steps to take to try to resolve the error.

Table 4. Error Symbols

ERROR SYMBOL	ERROR MESSAGE	DESCRIPTION
	Out-of-balance during centrifugation	The centrifuge has detected an imbalance during centrifugation and has stopped centrifugation. Open the cover and ensure that: - The head contains an even number of cards. - The cards are distributed symmetrically. - The cards are loaded properly in the card holders. - The DG SPIN head is not damaged (i.e., no dents, deep scratches, etc.). - The card holders are in their proper position. - The DG SPIN head is properly placed on the axle. After you have corrected the cause of the imbalance, close the cover, press the Start Centrifugation button , and make sure that the Out-of-Balance symbol  disappears and that centrifugation begins. If the error symbol appears again or if centrifugation does not start, contact Technical Service.

ERROR SYMBOL	ERROR MESSAGE	DESCRIPTION
	Erroneous centrifugation time	The centrifugation time of the batch in progress was different from the pre-programmed time of nine minutes. The timer flashes the time remaining and the alarm beeps. This error may be due to the following situations: - An operator has stopped the centrifugation batch as described in Section 6.3. - A power failure has stopped the centrifugation batch. Press any button to stop the alarm and reset the timer for a new centrifugation batch.
	The centrifuge cover is open	For safety, the centrifuge is designed so that centrifugation cannot start with the cover open. Ensure that the cover is properly closed. If the Cover Open symbol  still appears, contact Technical Service. If the cover is closed but the Cover Open symbol  flashes when the Start Centrifugation button  is pressed, make sure that no centrifugation batch remains in the centrifuge from the previous centrifugation batch.
	Centrifugation speed out-of-range	A speed error has been detected during centrifugation. Open the cover, remove the DG SPIN head, and discard the cards. Turn the centrifuge off and back on again. If the error symbol appears again, contact Technical Service.

ERROR SYMBOL	ERROR MESSAGE	DESCRIPTION
	Contact Technical Service	If the centrifuge detects any alteration in its function that cannot be resolved by the operator, the Repair symbol  appears. Turn the centrifuge off and back on again. If the error symbol appears again, contact Technical Service.
 	Motor overload	The centrifuge has detected motor overload during centrifugation. Centrifugation stops and the alarm beeps. Press the Start Centrifugation button  or the Stop Centrifugation button  to stop the alarm. Open the cover, remove the DG SPIN head, and discard the cards. Turn the centrifuge off and back on again. If the error symbols appear again, contact Technical Service.
  	The out-of-balance sensor does not work correctly	The centrifuge has detected an error in the out-of-balance control system during centrifugation. Centrifugation stops and the alarm beeps. Press the Start Centrifugation button  or the Stop Centrifugation button  to stop the alarm. Open the cover, remove the DG SPIN head, and discard the cards. Turn the centrifuge off and back on again. If the error symbols appear again, contact Technical Service.

ERROR SYMBOL	ERROR MESSAGE	DESCRIPTION
 	Cover opening error	The centrifuge has detected an error in the cover opening system. The Cover Open symbol  flashes and the alarm beeps. Press the Start Centrifugation button  or the Stop Centrifugation button  to stop the alarm. Turn off the centrifuge and unload the cards from inside using the security ring cover as described in Section 4.6. Turn the centrifuge on. If the error symbols appear again, contact Technical Service.

If the centrifuge detects any error conditions other than those errors already mentioned, the centrifuge will:

- Stop centrifugation.
- Not respond to the **Start Centrifugation** button , the **Stop Centrifugation** button , or the **Open Cover** button .
- Activate the alarm.
- Display the **Repair** symbol , along with an **E** and a corresponding error code.



CAUTION: If the DG SPIN Centrifuge displays an error code, turn the centrifuge off and then back on again. If the error symbols appear again, contact Technical Service and give them the error code.

12.2 Alarm Situations

The DG SPIN Centrifuge has an alarm with three volume level settings. The alarm sounds in the following situations:

- The centrifugation batch is finished. The alarm emits a long beep.
- A centrifugation batch has been stopped by pressing and holding the **Stop Centrifugation** button . The alarm emits three beeps and a long beep, and the timer flashes the time remaining.

- A centrifugation batch has been stopped because of a power outage. When the power comes back on, the alarm beeps and the timer flashes the time remaining from the time of the power outage.
- The cover has not been opened after a complete centrifugation batch and the **Start Centrifugation** button  is pressed. The alarm beeps and the **Cover Open** symbol  flashes. Open the cover and remove the cards.
- An error has occurred during centrifugation: imbalance, centrifugation speed out-of-range, malfunction of the out-of-balance control systems, fault in the cover opening system, overload of the motor, etc. The alarm beeps and the corresponding error symbol is displayed.



CAUTION: To stop the alarm, press the **Open Cover** button , the **Start Centrifugation** button , or the **Stop Centrifugation** button . If the alarm continues, contact Technical Service.

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Appendix A. Maintenance Log

Model: Serial No: Installation date:

DATE	ACTION CARRIED OUT	PERFORMED BY & DATE	SUPERVISOR & REVIEW DATE

Maintenance Log

DATE	ACTION CARRIED OUT	PERFORMED BY & DATE	SUPERVISOR & REVIEW DATE